

Machine Learning Fundamentals

Technical reference document for Retrieval-Augmented Generation (RAG) practice.

1. What Is Machine Learning?

Machine learning (ML) is a branch of artificial intelligence in which computer systems learn patterns from data and use those patterns to make predictions, classifications, or decisions. Instead of writing every rule explicitly, a developer provides data, an objective, and a learning algorithm.

A typical ML workflow includes collecting data, cleaning and transforming it, selecting features or representations, splitting data into training and evaluation sets, training a model, measuring performance, and deploying the model.

- A model is the learned representation of a relationship between inputs and outputs.
- Training adjusts model parameters to reduce an objective or loss function.
- Inference is the process of using a trained model on new data.

2. Supervised Learning

Supervised learning uses labeled examples. Each training example contains input features and a target value. The model learns a mapping from inputs to targets and is evaluated on data it did not use for fitting.

- Classification predicts a discrete category, such as spam versus not spam.
- Regression predicts a continuous value, such as house price.
- Common algorithms include linear regression, logistic regression, decision trees, random forests, and neural networks.

3. Unsupervised Learning

Unsupervised learning works with data that does not have an explicit target label. The goal is often to discover structure or useful representations in the data.

- Clustering groups similar examples.
- Dimensionality reduction represents data using fewer dimensions while attempting to preserve useful structure.
- Examples include k-means clustering and principal component analysis (PCA).

4. Training, Validation, and Test Sets

A dataset is commonly divided into training, validation, and test portions. The training set is used to learn parameters. The validation set helps select hyperparameters or compare model configurations. The test set is reserved for a final estimate of generalization performance.

- Data leakage occurs when information from outside the training process improperly influences training or model selection.
- Random splitting is not always appropriate for time series or grouped data; the split should match the deployment setting.

5. Overfitting and Underfitting

Overfitting occurs when a model fits the training data too closely and performs poorly on unseen data. Underfitting occurs when the model is too limited to capture important patterns.

- Regularization can reduce overfitting.
- More representative training data can improve generalization.
- Cross-validation can provide a more robust estimate when the dataset is limited.

6. Model Evaluation

Evaluation metrics should match the task. Accuracy is useful in some balanced classification problems, but it can be misleading when classes are imbalanced.

- Precision measures the fraction of predicted positives that are actually positive.
- Recall measures the fraction of actual positives that are identified.
- F1 score is the harmonic mean of precision and recall.
- Mean squared error and mean absolute error are common regression metrics.

7. Neural Networks

A neural network is composed of layers of parameterized transformations. A basic feed-forward network passes an input through one or more hidden layers and produces an output. Nonlinear activation functions allow the network to model complex relationships.

- Backpropagation computes gradients of the loss with respect to model parameters.
- Gradient-based optimization updates parameters to reduce the loss.
- Deep learning refers to neural networks with multiple learned layers.

8. ML and RAG

In a RAG application, machine learning concepts are relevant to embeddings, ranking, classification, and evaluation. An embedding model maps text to numerical vectors so semantically related content can be retrieved from a vector database.

- Chunk quality affects retrieval quality.
- Evaluation should test both retrieval relevance and final answer quality.

- A strong RAG system separates document ingestion, retrieval, generation, and evaluation so each component can be tested independently.